

Course: IS Strategy, Management, and Acquisition

Project Title

Design and Develop a Database Management System to Support Organizational Decision-Making

Project Objective

The objective of this project is to enable students to:

- Analyze organizational information requirements.
- Design a relational database.
- Implement the database using a DBMS.
- Demonstrate how databases support Information Systems Strategy and Management.
- Produce reports that assist managers in making strategic decisions.

Group Formation

- Each group must select **one organization** or business scenario.

Examples include:

1. Hospital
2. University
3. Hotel
4. Supermarket
5. Pharmacy
6. Bank
7. Library
8. Airline

Project Tasks

Task 1: Organization Analysis (10 Marks)

Describe the selected organization.

Include:

- Background
- Vision and mission
- Business processes
- Existing problems
- Why a database system is needed

Task 2: Requirements Analysis (10 Marks)

Identify:

- Users of the system
- Functional requirements
- Non-functional requirements

Task 3: Database Design (20 Marks)

Design the database.

Include:

Entities

Example

- Patient
- Doctor
- Appointment
- Department
- Payment

Attributes

Example

Patient

- PatientID
- Name
- Gender
- Address
- Phone

Relationships

Example

- One doctor treats many patients.
- One patient can have many appointments.

Task 4: ER Diagram (15 Marks)

Develop an Entity Relationship Diagram (ERD).

The ERD should include:

- Entities
- Relationships
- Primary Keys
- Foreign Keys
- Cardinality

Students may use:

- Draw.io
- Lucidchart
- Microsoft Visio
- MySQL Workbench

Task 5: Database Normalization (10 Marks)

Normalize the database up to **Third Normal Form (3NF)**.

Students should show:

- UNF
- 1NF
- 2NF
- 3NF

Explain each stage.

Task 6: Database Implementation (20 Marks)

Create the database using one of the following:

- MySQL
- SQL Server
- PostgreSQL
- Oracle Database

Students must:

- Create tables
- Define primary keys
- Define foreign keys
- Insert sample records (minimum 50 records across all tables)

Task 7: SQL Queries (15 Marks)

Write and execute SQL statements.

Include at least:

- 10 SELECT queries
- 5 UPDATE queries
- 5 DELETE queries
- 5 INSERT queries
- 3 JOIN queries
- 3 Aggregate queries
- 2 Nested queries

Task 8: Reports and Strategic Analysis (20 Marks)

Generate useful management reports.

Examples:

Hospital

- Number of patients treated
- Monthly revenue
- Most visited doctor

University

- Student enrollment
- Department performance

- Fee collection

Hotel

- Room occupancy
- Monthly income
- Customer statistics

Then explain:

- How these reports help management.
- How the database supports organizational strategy.
- How managers can use the information for decision-making.